Education Tech Research Dev
https://doi.org/10.1007/s11423-026-10601-6

RESEARCH ARTICLE

A meta-synthesis of automatic writing evaluation research: trends and developments over a decade

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Received: 28 April 2025 / Accepted: 17 February 2026

Abstract

Automated writing evaluation (AWE) technologies have emerged as promising tools that streamline the feedback process and strengthen students' writing skills. This meta-review synthesized eleven systematic reviews and meta-analyses on AWE research published from 2015 to 2025. Before the main analyses, all selected reviews were evaluated using Many-Facet Rasch Model (MFRM) to determine the study quality. Next, syntheses methods employed narrative approach and text mining analysis. The results suggested the shift from rule-based AWE system to AI-driven AWE tools over three decades. The synthesized findings from meta-analyses supported the effectiveness of AWE on surface-level writing (e.g., grammar, spelling) but highlighted its limitations in improving high-order level of writing (e.g., argumentation). Further, drawing on moderator analyses, educational levels and duration deserve attention in the implementation of AWE. Finally, persistent challenges, future research directions, and practical pedagogy were also identified and discussed. Overall, the present meta-synthesis study supports the potential value of AWE as an adjunct tool rather than a replacement for human feedback in writing instruction.

Keywords Automated writing evaluation (AWE) · Meta-Review · Many-Facet Rasch Model (MFRM) · AI-Driven tools · Feedback

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Published online: 04 March 2026

Introduction

Contemporary applications of automated writing evaluation (AWE) systems draw on advanced machine learning and large language models to establish analytical routines for writing evaluation. Expanding enrollments and rising expectations for timely feedback push many programs to look for scalable solutions. Ongoing advances in natural language processing widen the analytical scope of evaluation and bring into view discourse patterns, argumentative structure, and organization of content, which together support attention to multiple dimensions of composition quality. Advances in machine learning and large language models now drive systems to move past surface error detection and dig into rhetorical structure, coherence, and development of ideas.

In parallel, the body of empirical research on AWE also mushrooms, and a large number of systematic reviews and meta-analyses have been released to compile the growing body of AWE literature (Cheung, 2015; Zhang, 2021). Of note, the reviews have examined how well AWE works in different languages, for example, German (Shi & Aryadoust, 2024), English (Huang et al., 2024), and L1 (Chen et al., 2025). On the one hand, the majority of systematic reviews exclusively investigated teacher perspectives (Nunes et al., 2022) and student perceptions about the deployment of AWE tools in language classes (Cheung, 2015; Fu et al., 2024). On the other hand, the meta-analytic studies investigated possible moderator variables like educational levels (Chen et al., 2025; Fleckenstein et al., 2023), study duration (Chen et al., 2025), and feedback types (Xue, 2024). Although several reviews on AWE exist, fragmentation of knowledge persists across them.

Apart from established diverse findings, several issues have been highlighted by scholars and call for clarification, for example, long-term effectiveness, teacher–student perceptual alignment, and classroom enactment of AI-supported tools (Karatay & Karatay, 2024). Hence, in view of a dispersed scope of AWE reviews and mixed evidence, the purpose of the meta-synthesis is intended to address the issues through the first systematic integration of findings from systematic reviews and meta-analyses that span a full decade from 2015 to 2025.

To fill the research gap, this study conducted a meta-review of six systematic reviews and five meta-analyses on AWE to provide evidence evolution of AWE over three decades and highlight research areas with limited coverage. In specific, this overview of review pursues three main objectives: *(1)* mapping broader developments in AWE research, *(2)* examining overall effects and the role of contextual factors in AWE interventions, and *(3)* proposing directions for future inquiry grounded in evidence. To achieve, we adopted a combined narrative and text mining methods. The findings of this meta-review could provide a comprehensive understanding regarding the theoretical and pedagogical application of AI-driven writing support and thus offer a unified, accessible reference for researchers, educators, and policymakers. These objectives guide three following research questions:

Research question 1: What are the common trends and methodological strategies across existing systematic reviews and meta-analyses of AWE?
Research question 2: What is the overall effectiveness of AWE, and how do contextual factors shape the impact of AWE in language education?
Research question 3: What key gaps and themes for future research arise from the consolidated findings?

Literature review

Automatic writing evaluation

Chronologically, early tools of AWE had functions of automated proofreading, grammar and spelling correction, and detection of punctuation errors (Burstein, 2003). Later, AWE acted the role as a computerized system that can quickly as well as accurately diagnose a range of linguistic and structural components of writing (Li et al., 2015). As technology progressed further, tools were developed to evaluate stylistic features and guide students in lexical choices (Elliot, 2003), logical organization, and cohesion between paragraphs (Li et al., 2015). More recently, the functions of AWE became subtler and more delicate, and can analyze content relevance through semantic similarity (Fu et al., 2024) together with an ability to assess argumentation and appraise the reasoning of writing performance (Wang & Gayed, 2024).

Pedagogical expectations attached to AWE have called forth numerous empirical projects investigating how AWE links to writing proficiency of learners. Zhang and Chen (2022) and Wilson and Czik (2016) report positive effects on cognitive development of students. Waer (2021) and Wilson and Czik (2016) also spell out gains in affective responses of learners. Other empirical work such as the investigation of Chen et al. (2016) points up distinctive strengths of feedback from teachers when intricate compositional facets come under review. A broad consensus in the field now holds the view AWE might as well serve as a complement to instruction guided by teachers rather than a replacement. In view of the conflicting research findings, and the continuing debate summarized by Wilson and Roscoe (2020), there remains a need for careful synthesis.

Previous review studies on the effectiveness of AWE

Systematic reviews

To date, several systematic reviews have been conducted to synthesize the impact of AWE on language performance. Researchers acknowledge that AWE can improve surface-level writing skills, namely grammar, mechanics, and coherence. Moreover, Karatay and Karatay (2024) found that AWE may help with lexical complexity and syntactic variety, meaning that its influence extends beyond basic writing mechanics. Nevertheless, scholars continue to debate whether AWE contributes to the development of more advanced writing abilities, such as organization and argumentation (Cheung, 2015; Shi & Aryadoust, 2024; Zhang, 2021). On the one hand, some researchers indicate that structured writing programs that integrate AWE can help students organize their ideas more effectively and construct stronger arguments (Nunes et al., 2022). On the other hand, other scholars asserted that AWE tends to prioritize technical accuracy (e.g., grammar, spelling) over rhetorical effectiveness, and fails to provide sufficient support for students to develop logical coherence (Cheung, 2015; Zhang, 2021).

Concerning the discourse-level challenges in AWE tools, the findings of the prior reviews emphasized the importance of teacher involvement. In the case of independent learning, students often struggle to make sense of vague or overly generic AWE feedback, which can cause frustration and skepticism (Zhang, 2021), and, in turn, results in little academic improvement (Cheung, 2015; Karatay & Karatay, 2024). To overcome the challenges, Nunes et al. (2022) and Shi and Aryadoust (2024) suggested that AWE is more effective when complemented by teacher feedback.

In addition, the previous systematic reviews also documented teachers' perception and students' opinions based on their experience of using AWE tools. Some teachers perceived AWEs as a useful tool for providing structured feedback and reducing their grading workload (Zhang, 2021); others, however, are still doubtful of its accuracy of automatic scoring and its capacity to evaluate deep meaning in student writing (Karatay & Karatay, 2024; Shi & Aryadoust, 2024). Echoing with teachers' voices, some learners appreciate the immediacy of feedback (e.g., grammar and mechanics) from AWE tools (Nunes et al., 2022), whereas others expressed concerns over their superficial suggestions, such as spelling corrections (Cheung, 2015; Zhang, 2021). To overcome the surface-level suggestions of conventional AWE systems, AI-driven AWE applications (e.g., ChatGPT, Gemini, Copilot) were introduced, which not only provide writing mechanical corrections but also be able to generate more context-aware feedback. Despite the advanced functions of the AI-assisted AWE systems, recent research has discussed several issues relating to their reliability, bias, and also students' over-reliance on automation (Karatay & Karatay, 2024).

Meta-analyses

It is acknowledged that although systematic reviews provide valuable knowledge relating to benefits and challenges in implementing AWE tools, there is an absence of statistical evidence. To complete the picture, meta-analyses, which quantify the related findings, can provide further empirical evidence for the effectiveness of AWE tools. An early meta-analysis of Ngo et al. (2022) examined 24 AWE studies and generated a medium effect size of $g=0.59$ ($k=85$). In the same line, Fleckenstein et al. (2023) and Xue (2024) reported moderate effect sizes of $g=0.55$ and $g=0.42$, respectively. However, Huang et al. (2024) found a higher magnitude of 0.67, whereas Chen et al. (2025) identified a smaller effect size of 0.39 ($k=21$).

In addition to the overall effect size report, meta-analyses also carried out moderator analysis to define the potential factors that influence the effects of AWE in language learning. First, educational level received great attention from language researcher. Generally, university students consistently showed remarkable benefits from the use of AWE tools (Fleckenstein, 2023, $g=0.58$; Ngo et al., 2022, $g=0.62$). More modest effects are otherwise observed for both junior and senior high school students: small-to-moderate effects were reported in a range from 0.25 to 0.50 (Fleckenstein et al., 2023; Ngo et al., 2022; Xue, 2024; Chen, 2025). Second, the duration of an intervention also appears critical, as it was suggested by Ngo et al. (2022) and Fleckenstein et al. (2023) that medium-term use of three to nine weeks brings about the greatest gains, but the prolonged exposure, in contrast, may lead to diminishing returns (Xue, 2024). Also, L2 learners appear to gain more than L1 learners (Fleckenstein et al., 2023; Xue, 2024).

The existing systematic reviews and meta-analyses appear to present only the fragmented and disconnected lines and dots of the full landscape: *(1)* although the prior reviews have reached agreement on the positive effects of AWE on language learning, their divergent scopes, outcome definitions, and methods temper the ease of comparison; *(2)* meta-analyses quantify effect sizes but do not always account for contextual variations in implementation; systematic reviews provide qualitative insights but often lack comprehensive comparisons across studies. Given the above limitations, a review of reviews synthesizes broader existing evidence (i.e., qualitative content and statistical values) to map out consistencies and contradictions, which possibly offer a more structured understanding of the complexities surrounding AWE research. The results of the present meta-review, in turn, could inform future studies by clarifying the extent to which AWE contributes to writing development and under what conditions it is most effective.

Method

The method follows Systematic Meta-Review Guidelines from Hennessy et al. (2019). In specific, the procedure includes scope specification, data sources, screening strategies, synthesis methods, and finding reports (Fig. 1).

Define the scope

The aim of this step is to establish a clear research focus that only includes the review papers synthesizing the effects of AWE on writing performance. To achieve, four guided questions were formulated: *(a)* When is the literature on AWE in language learning suitable for a meta-review? *(b)* What defines a strong meta-review topic in AWE research? *(c)*

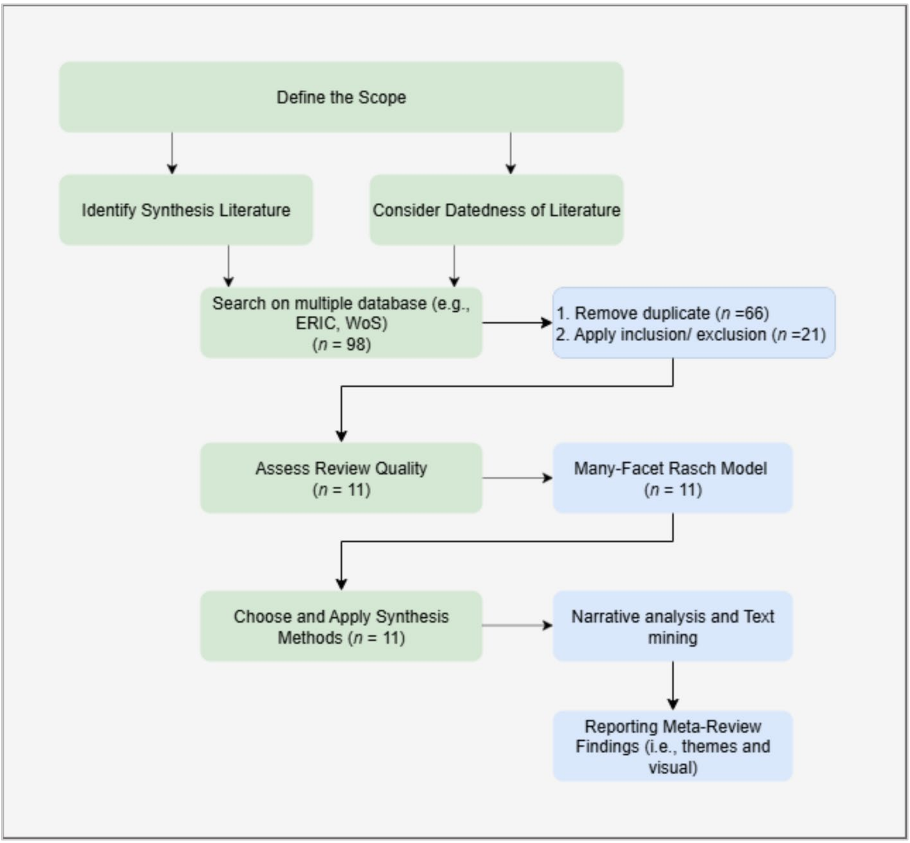


Fig. 1 Flowchart of data extraction

What have prior meta-reviews on AWE done well, where have they fallen short, and what gaps remain? (d) What resources and expertise are available for conducting a meta-review on AWE?. Thereafter, a set of keywords were identified and carried forward to the next step, which involved data searching.

Identify synthesis literature and datedness of literature

We searched relevant articles on different academic databases (i.e., ERIC, Web of Science, Google Scholar, SCOPUS, PsyINFO, ProQuest, ScienceDirect, and Education Source) using Boolean operators that combine four keyword categories: *(a)* Technology-related terms (e.g., "*computer-assisted*," "*AI-generated feedback*"); *(b)* Type of feedback (e.g., "*automated evaluation*", "*AI-based assessment*"); *(c)* Writing performance outcomes (e.g., "*writing performance*", "*writing achievement*"); and *(d)* Research approach (e.g., "*systematic review*", "*meta-analysis*"). To maximize the scope, we impose no publication date restrictions.

Remove duplicate

Thereafter, the title and abstract of the initially included studies were carefully screened to detect overlapping studies retrieved. This step ensures that each included study provides unique and non-redundant evidence by systematically identifying and eliminating duplicate records. Next, the process of screening study selection went through four predefined criteria. First, a selected study must be review articles, which could be systematic review and/or meta-analysis. Second, we collected peer-reviewed articles to ensure the high quality. Third, due to the proficiency limitation of researchers, we only synthesize peer-review articles that were written in English or Chinese. Finally, the included studies must report findings regarding the effectiveness of AWE in language learning.

Assess review quality

Methodological foundation of MFRM

In the present meta-review, we utilized the Many-Facet Rasch Model (MFRM) to evaluate the quality of the eleven selected reviews (i.e., 6 systematic reviews and 5 meta-analyses). The MFRM is an advanced psychometric framework derived from Item Response Theory and was originally put forward by Linacre (1989). The MFRM decomposes ratings into three components: *(1)* the quality of each review, *(2)* the severity of individual raters, and *(3)* the difficulty of checklist items – identify which items are harder to satisfy than others. Notably, the 27-item checklist developed from PRISMA was adopted to assess the study quality of the 11 systematic reviews and meta-analyses (Appendix A). Two researchers independently rated each review using the checklist items. Ratings are then transformed through a logarithmic procedure into interval-scale measures on a common logit metric, which permits comparison across all facets. In addition, fit statistics were computed to recover scoring irregularities, for example, aberrant rater behaviors or problematic items. Instead of exclusive dependence on agreement rates, the MFRM offers a more evidence-driven strategy to assess the quality of research reviews.

Table 1 MFRM analysis of the rating fit and consistency between raters

RATER PERFORMANCE SUMMARY

Rater	Outfit MNSQ	Fit Classification		Severity Profile	Rating Consistency	Distribution Statistics		
		Classification	Range			Mean Score	SD	Range
RaterA	1.326	Productive of measurement	0.5-2.0	Slightly more severe +0.024	Excellent (r=0.97)	1.098	0.376	0.259-1.481
RaterB	1.311	Productive of measurement	0.5-2.0	Slightly more lenient -0.024	Excellent (r=0.97)	1.097	0.362	0.296-1.481

Table 2 Study quality assessment by the MFRM analysis

STUDY-BY-STUDY QUALITY ASSESSMENT

Study	No. of Primary Studies	RaterA Score	RaterB Score	Difference (A - B)	Mean Logit Value	Outfit MNSQ	Agreement	Most Severe Rater	Study Quality Rank
Chen (2025)	25	1.481	1.481	0.000	1.481	1.399	Perfect	Equal	1st (Highest)
Nunes (2022)	8	1.444	1.407	+0.037	1.426	1.460	High	RaterA	2nd
Huang (2024)	22	1.370	1.370	0.000	1.370	1.418	Perfect	Equal	3rd
Fleckenstein (2023)	20	1.333	1.333	0.000	1.333	1.474	Perfect	Equal	4th
Xue (2024)	29	1.185	1.185	0.000	1.185	1.472	Perfect	Equal	5th
Ngo (2022)	44	1.148	1.111	+0.037	1.130	1.554	High	RaterA	6th
Shi (2024)	83	1.111	1.111	0.000	1.111	1.582	Perfect	Equal	7th
Karatay (2024)	40	1.074	1.074	0.000	1.074	1.702	Perfect	Equal	8th
Fu (2024)	48	0.926	0.926	0.000	0.926	1.804	Perfect	Equal	9th
Cheung (2015)	4	0.519	0.556	-0.037	0.538	1.890	High	RaterB	10th
Zhang (2021)	*0*	0.259	0.296	-0.037	0.278	1.940	High	RaterB	11th (Lowest)

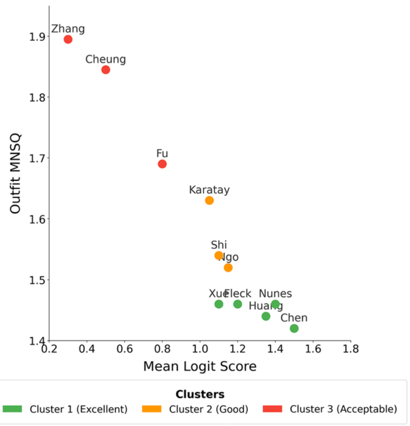
Fit Interpretation Guide

- 0.5 - 1.5: Productive of measurement (Ideal range)
- 1.5 - 2.0: Noticeable misfit - neither constructs nor degrades
- > 2.0: Severe misfit - degrades measurement
- < 0.5: Overly predictable (rarely of concern for persons)

Assessment results

Study quality and rater reliability As displayed in Tables 1 and 2 respectively, MFRM analysis provided a quality profile with independent ratings from two raters and the outfit mean square (MNSQ). The assessment revealed excellent inter-rater reliability since both raters reached a high rating consistency of 0.97 and showed productive measurement properties. Rater A scored an Outfit MNSQ of 1.326 and possessed a slightly more severe profile (+0.024), whereas Rater B scored an Outfit MNSQ of 1.311 with a slightly less harsh tendency (−0.024). Such minimal rater severity differences, combined with their high consistency ratings, validate the robustness of the quality assessment process. Both raters fell well within the productive measurement range, which confirms that the evaluation process remained free from systematic bias and measurement distortion.

Table 2 reports the quality assessment for all the included systematic review/meta-analysis study. MNSQ values ranged from 1.399 (Chen et al., 2025) to 1.940 (Zhang, 2021) and fell inside the productive range of 0.5 to 2.0 (Rasch convention). Perfect agreement was observed in seven of eleven studies. For the remaining four studies, the largest difference between raters reached 0.037 scale points. Agreement at such a level may support the



Cluster	Quality Level	Mean Logit Range	Outfit MNSQ Range
Cluster 1	Excellent	1.185 - 1.481	1.399 - 1.474
Cluster 2	Good	1.074 - 1.130	1.554 - 1.702
Cluster 3	Acceptable	0.278 - 0.926	1.804 - 1.940

Cluster	Cluster Characteristics
Cluster 1	High methodological quality with optimal fit statistics
Cluster 2	Good quality with acceptable but noticeable misfit
Cluster 3	Acceptable quality approaching caution threshold

Fig. 2 Cluster analysis of study quality

credibility of the quality stratification. Five reviews (i.e., Fleckenstein et al., 2023; Xue, 2024; Huang et al., 2024; Chen et al., 2025; and Nunes et al., 2022) reported MNSQ values from 1.399 to 1.554, which point to near-ideal fit (values close to 1.0). The remaining six reviews came in from 1.554 to 1.940 and still met accepted standards, although misfit signals did show up.

Assessment patterns of the raters suggested a consistent trend across reviews in the lower-quality tier. Rater B applied somewhat stricter criteria for the earliest publications, whereas Rater A appeared more lenient. The results for two early publications also flag up high MNSQ values and more limited empirical foundations. Zhang (2021) did not report the number of primary studies and recorded an MNSQ of 1.940, whereas Cheung (2015) included only four primary studies and recorded an MNSQ of 1.890. The pattern indicated that early reviews in the field of AWE operated under less developed methodological designs and thinner evidence bases compared to later syntheses. Even so, initial reviews add historical context and help set the stage for more comprehensive syntheses which followed. For the same reason, inclusion of early work in the meta-synthesis seems warranted since the record helps trace the evolution of methodology in AWE across the past decade. Two raters' (the leading author and the 3rd author) evaluation also appears to help hold up methodological soundness and maintain independence from any single rater.

Quality stratification Quality stratification was performed using cluster analysis to investigate how the 11 selected reviews are positioned based on their Mean Logit Scores and Outfit MNSQ values. Both K-means clustering and hierarchical clustering with single linkage showed consistent three-cluster solutions. The results of cluster analysis of study quality (Fig. 2) showed that reviews with the highest Mean Logit Scores (1.185–1.481) and lowest Outfit MNSQ values (1.399–1.474) (i.e., Chen et al., 2025; Fleckenstein et al., 2023; Huang et al., 2024; Nunes et al., 2022; Xue, 2024) grouped together to present "Excellent Quality" cluster. Further, the hierarchical clustering showed that Cluster 1 achieved the strongest inter-study similarity. In comparison to cluster 1, Cluster 2 (Good Quality) contained reviews that showed lower Mean Logit Scores (1.074–1.130) and higher Outfit MNSQ values (1.554–1.702). Lastly, Acceptable Quality Cluster 3 grouped reviews which hardly met the inclusion criteria with the lowest Mean Logit Scores (0.278–0.926) and the highest misfit values (1.804–1.940), yet which were still valuable in presenting preliminary comment on the development of AWE research. Simply put, as methodological rigor increases, measurement misfit tends to go down.

In summary, the inter-rater reliability reached $r = 0.97$ and rater bias remained minimal with a ± 0.024 MNSQ difference. Sixty-four percent of studies showed perfect agreement and the remainder showed a maximum discrepancy of 0.037 scale points. All eleven studies fell within the productive measurement range with MNSQ values within the accepted threshold of 2.0. An inverse relationship between mean logit scores and outfit MNSQ values indicated internal consistency of the quality structure. Based on the valid results of MFRM, five meta-analyses and six systematic reviews were convincingly included in the current meta-review. Key characteristics of the selected reviews were presented in Table 3

Given the positive results of the quality assessment, the corpus from the eleven qualified reviews is well suited for text mining and data analytics to identify trends and patterns. The corpus drawn from the eleven qualified reviews is well suited for text mining and data analytics because it synthesizes evidence from more than 300 primary studies over a period of two decades. A coherent base for the robust detection of trends and evolutionary trajectories in AWE is constituted by the combination of recent high-quality evidence and foundational studies.

Synthesis methods and meta-review reports

Concerning the synthesis method, two researchers independently coded eleven meta-analyses and systematic reviews through a three-phase NVivo procedure, which is grounded in the thematic analysis methodology of Clarke and Braun (2014). The procedure was clearly presented in Fig. 3. In specific, we first examined each paper line by line and identified 142 unique codes with a mean distribution of 12.9 codes per study. Next, we engaged in 18 h of consensus meetings to identify patterns and redundancies across the initial coding set. As a result, systematic consolidation reduced our initial 142 codes to 87 refined codes, which represents a 61.3% retention rate that aligns with consolidation patterns reported in the qualitative synthesis literature (Guest et al., 2012). For example, when four separate studies characterized AWE feedback using identical descriptors such as "*abstract, vague, unspecific, formulaic, and repetitive*", these semantically equivalent observations were consolidated into a single, more robust code labeled "AWE feedback quality limitations". Finally, we employed higher-order thematic categorization to identify natural clustering patterns within the refined code set. Through iterative categorization cycles, we classified these 87 codes under five domains and each category ranged from 15 to 23 codes per category ($M = 17.4$, $SD = 3.36$). In order to provide evidence for the methodological coding rigour, inter-rater reliability was computed and the results showed a substantial agreement with a κ of 0.84 and an ICC [2, 2] of 0.86. In short, based on such systematically coding procedure, the qualitative codes in this meta-review are grounded in coded evidence rather than subjective interpretation.

All identified codes were then exported to Python, where a spaCy-based text-mining script computed *(1)* temporal frequencies and *(2)* co-occurrence matrix. First, annual frequencies for the 18 technological evolution codes were plotted and subjected to a timeline chart that shows generational phases and traces the progression of the field. Second, period frequencies for all 5 main themes were visualised in a diverging bar chart in which bar length denotes prevalence and bar colour indicates direction of

Table 3 Key information from the eleven included systematic reviews and meta-analyses

No	Study	Type	No. primary studies	Period	AWE tool	Language context	Main findings
1	Chen et al. (2025)	Meta-analysis	25	2012–2024	Not specific	L1/ L2	1. Overall performance: $g = 0.39$ 2. Significant moderator: Duration, Educational level
2	Huang et al. (2024)	Meta-analysis	22	2005–2023	Not specific	ESL	1. Writing performance: $g = 0.67$ 2. Affective outcomes: Motivation ($g = 0.87$); Anxiety ($g = -0.8$)
3	Xue (2024)	Meta-analysis	29	2008–2023	Criterion, Grammarly, Pigai, Ilias, Cambridge Write & Improve, CorrectEnglish	EFL/ ESL/ L1 as English	1. Writing performance: $d = 0.422$ 2. Significant moderator: Feedback type, Duration, Educational level, Genres of writing
4	Fu et al. (2024)	Systematic review	48	1993–2021	IntelliMetric, E-rater, Intelligent Essay Assessor, Pigai, iTEST, iWrite, Criterion, My Access, Writing Roadmap, Write to Learn	EFL/ ESL	1. AWE can reduce teacher workload; however, AWE primarily focus on surface-level error, but lack deep comprehension of meaning 2. Students' perception: quick feedback but inconsistent scoring
5	Shi and Aryadoust (2024)	Systematic review	83	1993–2022	Pigai, Criterion, Grammarly, Writing Pal, My Access, NC Write, Write & Improve, Microsoft Word, and newer AI-based tools	English, German	1. AWE is positive for surface-level writing performance (e.g., grammar, spelling) 2. AWE showed limited impact on deep writing aspects (e.g., coherence, argumentation) 3. AWE alone is less affective when it combines with teacher feedback

Table 3 (continued)

No	Study	Type	No. primary studies	Period	AWE tool	Language context	Main findings
6	Karatay and Karatay (2024)	Systematic review	40	2013–2021	Criterion, Pigai, Grammarly, MyAccess!, Writing Pal, Writing Mentor, Research-based tools	EFL/ ESL	1. AWE helps with surface-level (e.g., grammar, mechanics) 2. AWE has limited impact on higher-order writing skills such as argumentation and discourse organization
7	Fleckenstein et al. (2023)	Meta-analysis	20	Up to 2022	AI-driven feedback but not specific	L1/L2	1. Writing performance: $g=0.55$ 2. Significant moderators: Educational level, Duration
8	Ngo et al. (2022)	Meta-analysis	44	1993–2021	Criterion, Grammarly, Pigai	EFL/ ESL	1. Writing performance: $g=0.59$ 2. Significant moderators: Tools, Duration, Feedback target
9	Nunes et al. (2022)	Systematic review	8	2000–2020	PEG Writing, NC Write, Writing Roadmap, EssayCritic, Criterion, Summary Street	EFL/ ESL	1. AWE systems are more effective with teacher support 2. Teacher' perception: concerns about over-reliance on AWE

Table 3 (continued)

No	Study	Type	No. primary studies	Period	AWE tool	Language context	Main findings
10	Zhang (2021)	Systematic review	NA	NA	IntelliMetric, E-rater, Intelligent Essay Assessor, Pigai, iTEST, iWrite, My Access, Criterion, Holt Online Essay Scoring, Writing Roadmap, Write to Learn	General academic writing, Chinese EFL learners	1. AWE improves grading efficiency, provides structured feedback, and supports revisions 2. Teachers experience reduced workload but express concerns about over-reliance on algorithms 3. Students gain autonomy in writing but struggle with interpreting automated feedback 4. Feedback lacks contextual depth, may lead to test-oriented writing, and requires careful integration into teaching
11	Cheung (2015)	Systematic review	4	2008–2013	E-rater, IntelliMetric, Criterion, My Access	EFL/ ESL	1. Students' perception: negative, distrust 2. Teacher's perception: lack reliability

No Number, *NA* Not available, *AWE* Automated Written Evaluation, *EFL* English as a Foreign Language, *ESL* English as a Second Language, *L1* First language, *L2* Second language

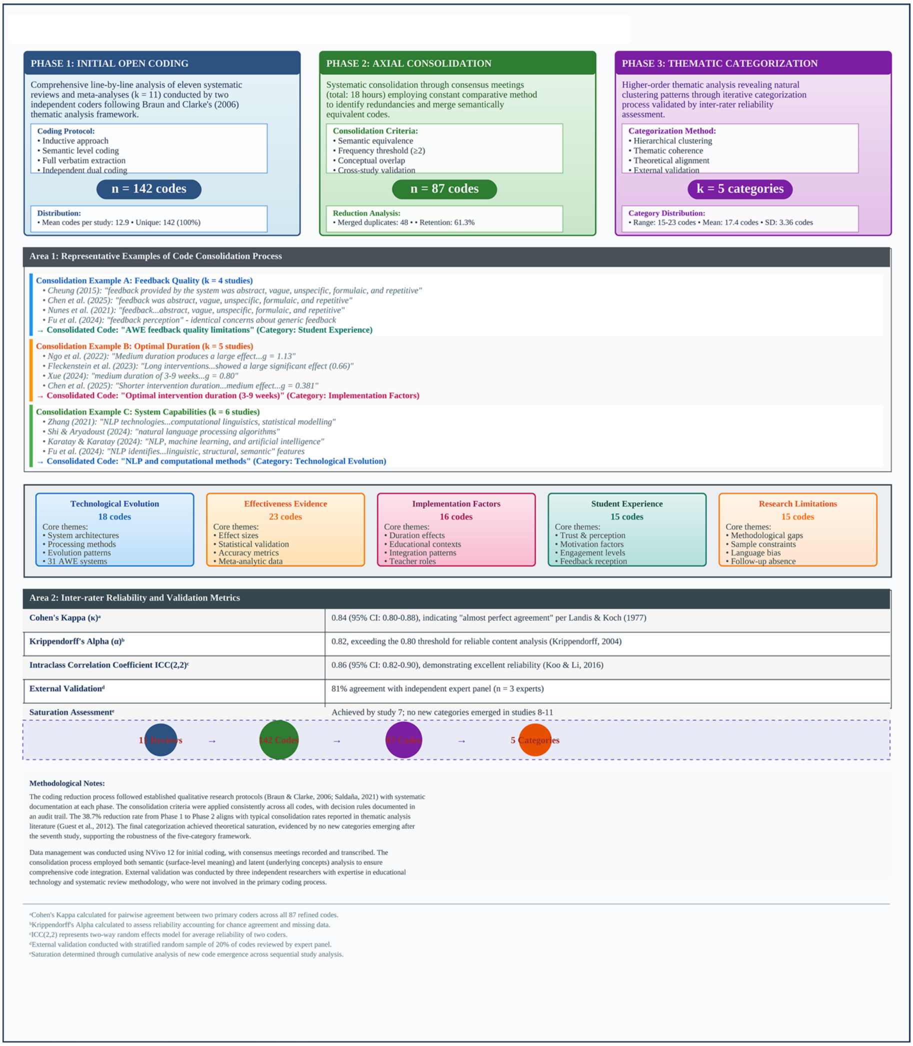


Fig. 3 Systematic consolidation process from initial open coding to thematic categories

change. Third, the study set up a co-occurrence matrix linking fifteen limitation codes to a set of solution codes derived from implementation factors and evidence of effectiveness. Rescaling of the matrix yielded a five-point priority surface. Two complementary visuals came out of the matrix. A bubble chart mapped out gap prevalence across two periods. Bubble size tallied total mentions and different colors represents the distinct gap status as persistent, emerging, or partially addressed. A heat map then turned the pairings between research gaps and future directions into priority scores. A score of five referred to the most frequent intersections and signaled the highest priority. The Appendix B lays out worked examples which show how specific codes from the 87-code framework fed into each analysis and the related visuals.

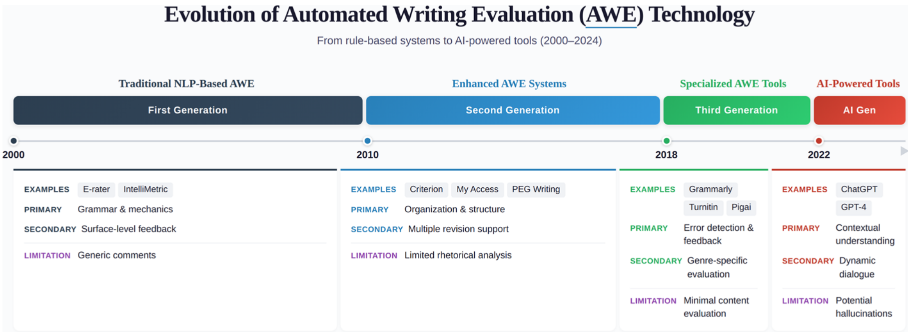


Fig. 4 Evolution of AWE from 2000 to 2024

Results

Methodological approaches and trends

Methodological approaches

The current meta-review comprised of 11 related reviews, including six systematic reviews, and five meta-analyses. In total, the 11 selected reviews synthesized 323 primary studies that examined the effectiveness of AWE on language learning. The majority of reviews applied the AMSTAR procedure or PRISMA criteria to assess the quality of their included primary studies, though with differing levels of rigor. With respect to the research scopes, systematic reviews (e.g., Karatay & Karatay, 2024; Nunes et al., 2022; Shi & Aryadoust, 2024) focused on user-oriented perspectives, namely students' opinions and teachers' beliefs. In comparison, the meta-analyses (e.g., Chen et al., 2025; Xue, 2024) tend to generate statistical evidence for overall effect sizes and their variations diverse learning contexts (e.g., learner population, duration of AWE use, and AWE tool types). Thus, although systematic reviews and meta-analyses differ in focus, both reinforce the evidence base by presenting distinct yet converging perspectives on AWE effectiveness.

Evolution of AWE research

As shown in Fig. 4, a chronological evolution of AWE technology presented four distinct generations from 2000 to 2024. The first generation (2000–2010) features "traditional NLP-based" systems such as E-rater and IntelliMetric, which concentrated primarily on grammar mechanics with surface-level feedback, though they possibly produced somewhat generic comments (Cheung, 2015; Fleckenstein et al., 2023; Ngo et al., 2022; Zhang, 2021). Moving forward, the second generation (2010–2018) introduced "Enhanced AWE Systems" (e.g., Criterion and PEG Writing) that have expanded into organization and structure assessment with apparent support for multiple revisions and able to provide rather limited rhetorical analysis (Huang et al., 2024; Nunes et al., 2022; Xue, 2024; Zhang, 2021). As technology evolved, the third generation tools (2018–2022), including Grammarly and Turnitin, conceivably specialized in error detection and genre-specific evaluation; however, they have experienced challenges with content evaluation to some extent (Fleckenstein et al., 2023; Shi & Aryadoust, 2024;

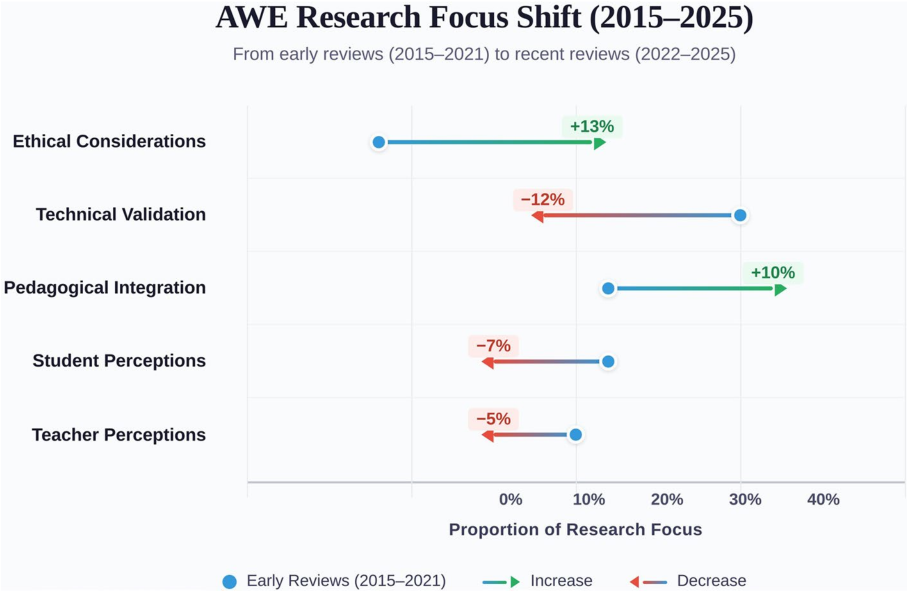


Fig. 5 Transitional shift (blue dot for early reviews and green dot for recent reviews)

Xue, 2024). In the current stage, AI-driven feedback emerged as the fourth generation (2022–2024) that integrate large language models (LLMs) for writing assessment. In comparison to earlier AWE generations, the AI-driven feedback systems are distinguished by their generative capabilities and better contextual understanding using either direct LLM applications (e.g., ChatGPT, Microsoft Copilot) or platform-integrated solutions (e.g., AI-enhanced Grammarly, Writing Pal). As a result, these AI-driven systems can provide narrative feedback that moves beyond template-based responses. According to Chen et al. (2025), the capability of generating such feedback appears to stem from an end-to-end architecture in which transformer-based models directly process the source text and generate assessment results. Regardless the significant effect sizes for narrative feedback ($g = 0.419$) (Chen et al., 2025), the extent to which LLM-native tools have been systematically evaluated seems relatively scarce in the literature.

Transformational shift

Figure 5 illustrates a transformational shift in the research priorities for studies of AWE between 2015 and 2024, where questions of functionality have yielded to analyses of implementation and impact. Ethical considerations (+13%) have received great attention from language researchers and pedagogical integration followed closely (+10%). In contrast, technical validation of AWE systems showed the steepest drop (−12%). Less attention was given to student opinions (-7%) and teacher perceptions (−5%) in recent years. The overall pattern signals that questions of functionality have yielded to analyses of implementation and impact, particularly in classroom practice and ethics, in step with AI-driven AWE exemplified by GenAI.

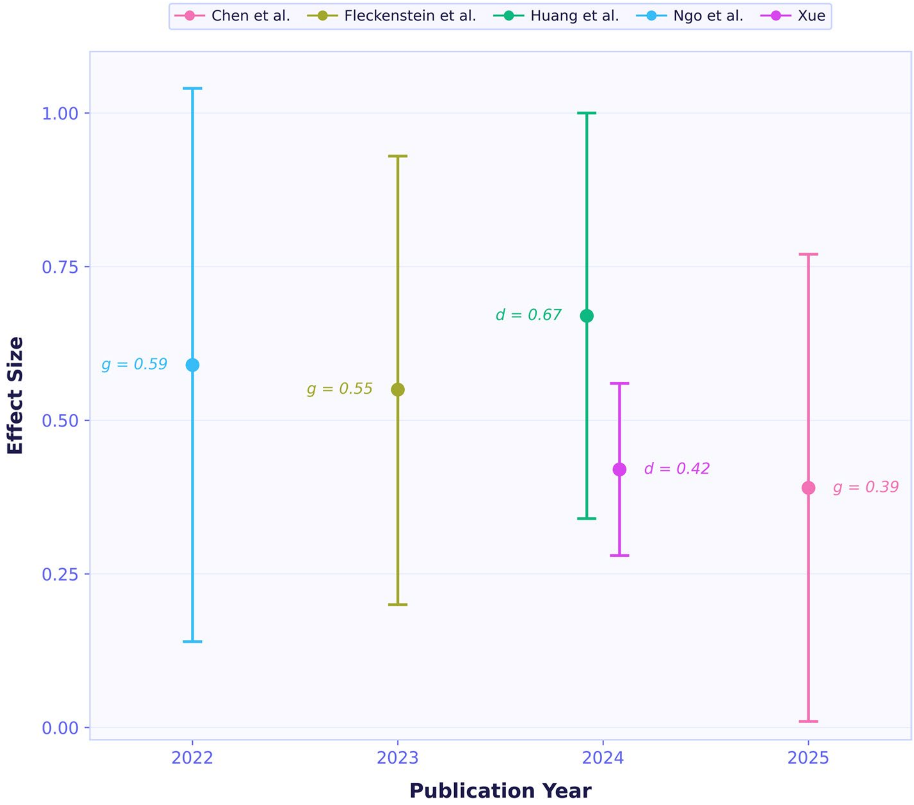


Fig. 6 Effect sizes of AWE across selected meta-analyses

The overall effectiveness of AWE on language learning

Both systematic reviews and meta-analyses indicated significantly positive effects of AWE on language performance. As shown in Fig. 6, Ngo et al. (2022) and Fleckenstein et al. (2023) found almost identically moderate effect sizes (Ngo et al., 2022: $g=0.59$, 95% CI = [0.15, 1.04]; Fleckenstein et al., 2023: $g=0.55$, 95% CI = [0.19, 0.91]). However, more recent analyses reveal increased variability. Huang et al. (2024) identified a moderate-to-large effect size ($d=0.67$, 95% CI = [0.34, 1.00]), whereas Xue (2024) reported a small-to-medium effect ($d=0.422$, 95% CI = [0.278, 0.565]). Most notably, Chen et al. (2025) investigated the effects of the AI-driven tools on language learning and reported the smallest effect size ($g=0.39$, 95% CI = [0.009, 0.770]). Despite the ability of automatically generated feedback, such small effect size reported by Chen et al. (2025) warrants caution about how AI-driven AWE systems should be used to improve language achievement.

The potential moderators

A growing body of literature suggests that AWE is more beneficial for students in higher education. Evidently, the related meta-analyses showed that adolescent students (i.e., junior high and senior high) obtained small-to-moderate effects of 0.25 and 0.50 (Fleckenstein et al., 2023; Ngo et al., 2022), whereas university students exhibited higher effect sizes (Fleckenstein et al., 2023, $g=0.580$, and Ngo et al., 2022, $g=0.620$) (Fig. 7). To further clarify the meta-analytic results, university students are frequently characterized in systematic reviews as more autonomous during feedback processing, whereas full benefit from AWE systems is often achieved by primary and secondary learners only when scaffolding of teachers is provided (Cheung, 2015; Fu et al., 2024; Zhang, 2021).

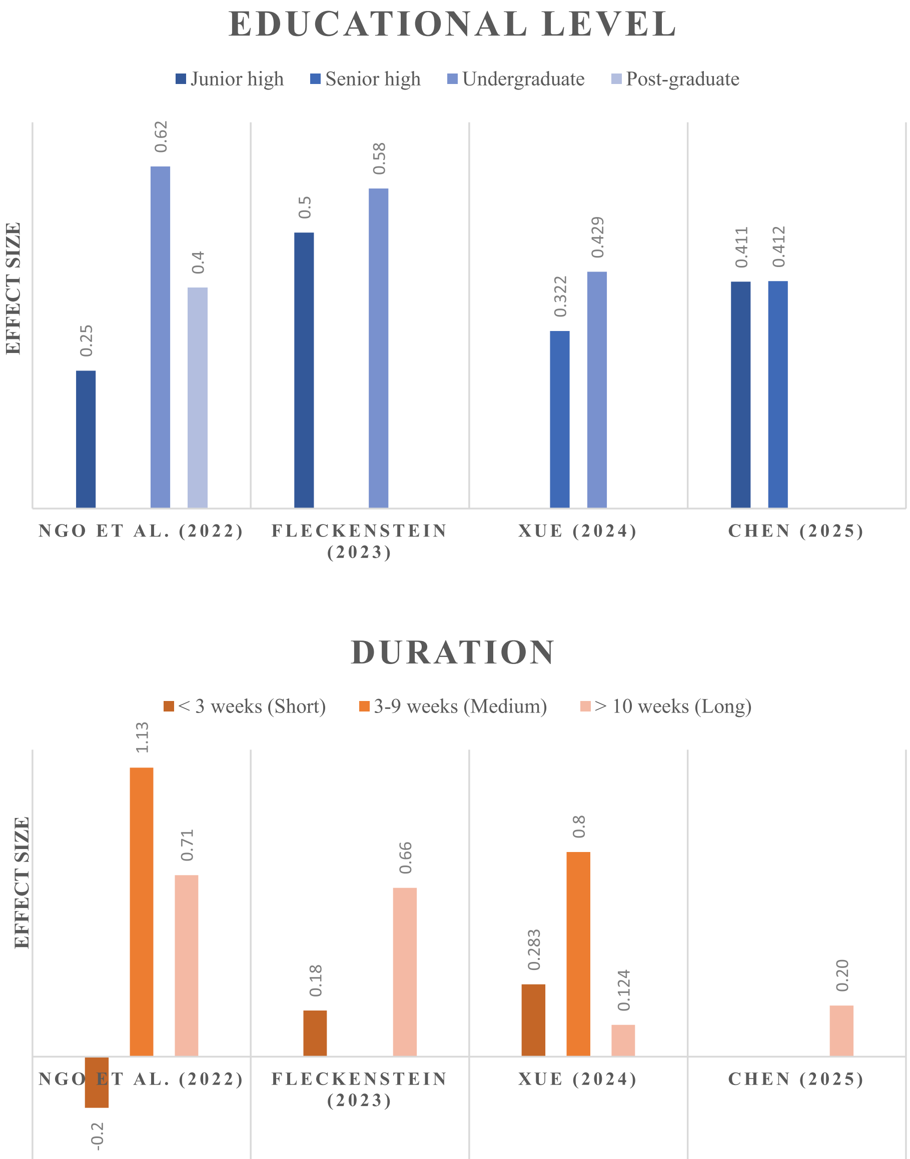


Fig. 7 Common moderators in AWE research

The duration of implementation also significantly modulates the effectiveness of AWE on language learning. Fu et al. (2024) and Shi and Aryadoust (2024) concurred that short-term exposure does not lead to significant improvement, but excessively long treatment results in feedback fatigue and decreased motivation. The meta-analytic findings also reported that short-term interventions (<3 weeks) obtained small benefits (Fleckenstein et al., 2023: $g = 0.18$, and Xue, 2024: $g = 0.283$) and even negative impacts (Ngo et al., 2022, $g = -0.20$). The effectiveness of AWE significantly increases when the duration was in 3–9-week period (Ngo et al., 2022: $g = 1.13$; Xue, 2024, $g = 0.80$). When durations beyond 10 weeks, the effects of AWE were inconsistent. Specifically, Fleckenstein et al. (2023) and Ngo et al. (2022) reported moderate-to-large effect sizes between 0.66 and 0.71, whereas other researchers (Chen et al., 2025; Xue, 2024) identified small effect sizes of 0.124 and 0.200, respectively.

Finally, AWE tools proved helpful in vocabulary acquisition (Cheung, 2015; Nunes et al., 2022), but showed little success in improving argumentation and coherence (Fleckenstein et al., 2023; Huang et al., 2024). In accordance with these qualitative findings, the meta-analyses indicated that the use of AWE tools remarkably influence on students' vocabulary acquisition (Ngo et al., 2022, $g = 0.83$), but show negligible advancement in supporting students' higher-order writing skills (Xue, 2024, $g = 0.227$). Overall, the AWE tools are useful for linguistic improvements, but its impact on higher-order writing need further evidence.

Persistent limitations and challenges

Figure 8 illustrates research gaps in AWE systems identified in systematic reviews (2015–2024). The identified gaps are grouped into three categories: Persistent (blue), Emerging (red), and Partially Addressed (purple). The size of the nodes reflects the frequency of each research gap across reviews, meaning that higher frequency resulted in bigger nodes.

Persistent research gaps

The network diagram suggests three persistent gaps in AWE research that have been noted across several reviews. The first gap discusses about student feedback, as many students appear to struggle with how they understand and how they use the feedback provided by AWE systems (Cheung, 2015; Nunes et al., 2022; Shi & Aryadoust, 2024; Zhang, 2021). The second gap concerns the long-term effects of AWE on language performance, which can be discovered through longitudinal studies (Nunes et al., 2022; Shi & Aryadoust, 2024). The third gap brings up whether AWE systems perform consistently across languages, dialects, and different cultural writing practices (Karatay & Karatay, 2024; Shi & Aryadoust, 2024).

Emerging research gaps

AI-driven AWE tools have recently introduced novel research concerns. The visualization highlights three emerging gaps, including LLM Transparency, AI Ethics and Privacy, and Algorithmic Bias. The three gaps illustrate significant changes brought about by LLMs in AWE technology with respect to assessment decision processes, data protection protocols, and bias perpetuation (Shi & Aryadoust, 2024).

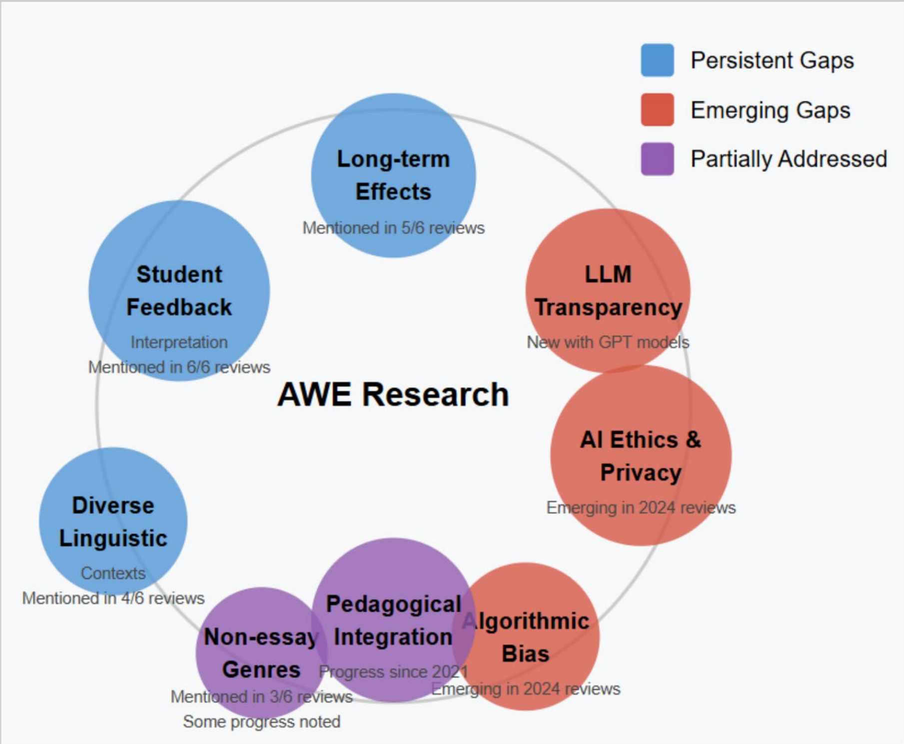


Fig. 8 Identified research gaps

Partially addressed gaps

Concerning the partially addressed gaps, our findings indicated two major gaps, which are non-essay genres and pedagogical integration. Non-essay genres remain scant, with three reviews mentioning limited research on the applicability of AWE to diverse writing forms, for instance, reports or creative writing (Karatay & Karatay, 2024; Shi & Aryadoust, 2024; Zhang, 2021). In terms of pedagogical integration, Nunes et al. (2022) found a growing alignment between AWE and instructional practices. Nevertheless, they stress that challenges persist in optimizing these tools to effectively complement feedback from teachers.

Implication and future direction

The heatmap matrix (Fig. 9) points to important gaps potentially existing in the current research stage of AWE. In addition, findings suggest value in interdisciplinary approaches, process-oriented work, and comprehensive frameworks able to account for ethical, cultural, longitudinal and learner focused considerations. Implementation studies on institutional factors are also called for to assess the effectiveness of AWE in science subjects. Further, the cells marked “4” suggest the relevance of adopting multiple data collection methods (e.g., think-aloud, eye-tracking, revision analysis) to capture learning behaviors.

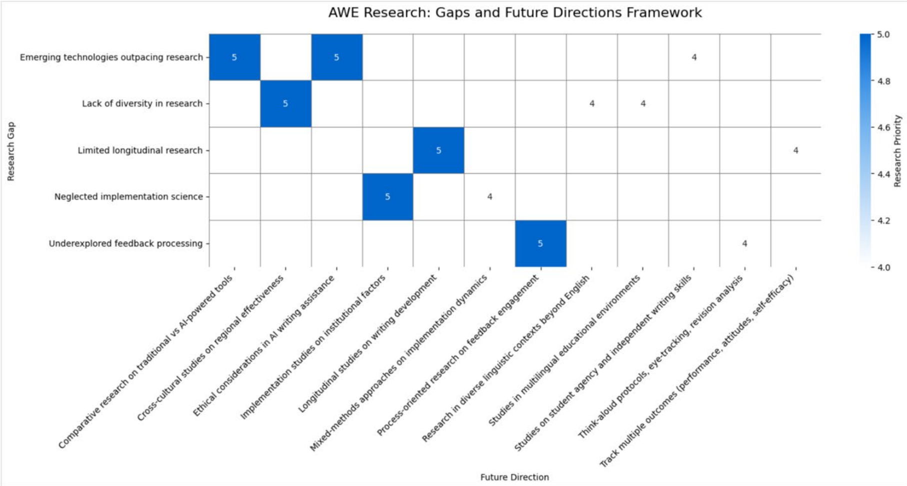


Fig. 9 The visualization of the heatmap

Discussion

Methodological innovation of the present meta-synthesis study closes a clear gap in research on AWE. Qualitative systematic reviews yield rich context but often struggle to generalize across settings. Quantitative meta-analyses focus on effect sizes and tend to strip out context. The present synthesis pulls together the two traditions, sets them side by side, and weaves their results into a single analytic frame. The approach lines up effect estimates and contextual explanations to form a coherent view of patterns of effectiveness of AWE. As a result of the implementation of newly-conceived methodological framework, the study spots research priorities that have grown in importance as technology has moved on. Ethical concerns appear in four of five recent reviews but only in two of six early reviews. Questions about cross-linguistic applicability have doubled from earlier to later reviews, although systematic work in that area remains thin. A cross-review synthesis makes the trends visible and sets the stage for targeted follow-ups.

Research evidence points to modest and inconsistent gains in higher-order writing skills such as argumentation, organization, and discourse coherence (Xue, 2024). The overall pattern signals system-level problems and calls for a sustained, evidence-based response rather than isolated one-off projects. Accordingly, a system-wide response should take shape. Professional learning of teachers should be phased in and tied to curriculum goals. Programs should build up feedback literacy of students through explicit instruction (particularly for young learners) and guided practice. Institutions should weave AWE into multi-cycle revision and assessment routines so learners can plan, draft, receive feedback, and rework text across several rounds. In sum, a coherent program should roll out across sites and keep alignment of tools, pedagogy, and assessment in view.

Evolution and trends in AWE research

As a whole, it can be seen, upon a thorough and comprehensive text mining cycle, that the attention in research on AWE has moved away from surface-level feedback tools toward AI-driven systems that generate more elaborate comments. It has been observed that AI-driven feedback at times drifts into generic remarks and misinformation, and concern about reliability has been raised in work of Huang et al. (2023). Given such risks, therefore, pedagogical integration and ethical governance have been proposed to sit at the centre of current debates as a critical response to the arrival as well as the massive applications to GenAI to language education. Because according to the academic convention upheld by Computer-Assisted Language Learning (CALL), the role of technology should be to supplement, rather than supplant, human instruction, the cultivation of students' writing skills still needs to be scaffolded by teacher's professional judgment and evaluation (Chapelle, 2003). Viewed in this light, we contend that future research should map the circumstances in which outputs of AI and judgment of the teacher combine so that collaboration between human and AI improves achievement of students.

Overall effectiveness of AWE on language learning

The synthesis of both meta-analyses and systematic reviews supports the positive effects of AWE on students' writing performance, and the mechanisms of AWE can account for the pattern. According to Zhang (2021) and Shi and Aryadoust (2024), immediate and real time feedback from AWE tools help create favourable conditions for continuous revision and in turn for later improvement in writing. AWE tools are also found capable of guiding learners toward more structured arguments and clearer written communication as a result of the newly added functions such as genre templates, discourse analysis, and coherence verification systems (Shi & Aryadoust, 2024). However, recent studies examining AI-driven systems report smaller effects. A potential explanation for the smaller effects may be attributed to the inconsistency in feedback from AI, a quality contingent upon the style of writing, the level of proficiency, and the linguistic background of students (Chen et al., 2025; Shi & Aryadoust, 2024). On account of the presentation of multiple revision options, a rise in cognitive demand may also occur. A subsequent increase in cognitive load can prompt lower-achieving learners to pull back from the writing process (Karatay & Karatay, 2024).

Most meta-analyses reported wide confidence intervals that signal a need for moderator analysis. In the synthesis it was found that educational level and experimental duration function as recurrent moderators. Researchers point out that undergraduate students tend to gain the most from AWE because many university cohorts are digitally fluent, and a TAM survey of 448 freshmen and sophomores confirmed high intention to use AWE, as reported by Zhai and Ma (2022) and He and Zhu (2017). Regarding duration, it was observed that effect size reached a peak between 3 and 9 weeks and that a motivational curve in which initial novelty gradually fades may account for the timing of the peak.

Research gaps and future direction

Comparative research on traditional vs. AI-driven tools

The emergence of large language models (e.g., chatGPT) has mounted a knowledge gap about their effectiveness over conventional AWE systems. Even with possible new frontiers in feedback quality, institutions currently lack empirical studies to inform decision-making regarding relative merit, cost-effectiveness, and pedagogical impact. Careful comparative research would support educators and administrators to invest their resources effectively, and apply the appropriate technologies for specific learning contexts, which in turn contributes to a higher quality of instruction.

Ethical concerns

Despite the promising effects, the use of AI-supported writing raises immediate ethical concerns regarding academic honesty, intellectual property, and authentic skill development (Shi & Aryadoust, 2024). Without clear guidelines for proper use and attribution, educational institutions risk the inadvertent promotion of plagiarism or the weakening of critical thinking. Chen et al. (2025) and Huang et al. (2024) argue that research into equity, transparency, and responsible implementation is therefore crucial to uphold educational values as AI capabilities in education continue to advance.

Cross-cultural studies on regional effectiveness

The research of AWE of English-speaking, Western environments is well-documented (Shi & Aryadoust, 2024). Nonetheless, few studies have been undertaken to investigate whether the effects of AWE tools may differ as a function of different languages, of which writing approaches and standards of non-English languages are distinct from English writing (Chen et al., 2025; Xue, 2024). To interpret, rhetorical customs, linguistic structures, and cultural expectations shape how learners respond to feedback (Huang et al., 2024). Hence, an expansion of inquiries into diverse settings can address equity concerns and determine adjustments needed for automated feedback systems (Fleckenstein et al., 2023). Such efforts encourage more inclusive solutions that align with a broader range of instructional contexts (Ngo et al., 2022).

Longitudinal studies on writing development

To date, the majority of AWE research focuses on short-term outcomes at the cost of the incremental growth that occurs in the development of written skills (Fleckenstein et al., 2023; Shi & Aryadoust, 2024). Notably, writing abilities take time to develop. However, there is little knowledge about the degree to which the achievement obtained via AWE tools at earlier time can be likewise translated into significant improvement at later time (Ngo et al., 2022; Xue, 2024). Thus, future research is suggested to conduct longitudinal studies that can track whether AWE interventions lead to stable gains or only short-term changes in language learning (Chen et al., 2025; Huang et al., 2024).

Implementation studies on institutional factors

Real-world adoption of AWE depends on institutional variables such as technology infrastructure, administrative support, faculty acceptance, and alignment with existing curricula (Shi & Aryadoust, 2024). Further, it is worth noting that if integration strategies are inadequate or stakeholders resist change, sophisticated AWE tools may still prove ineffective. In other words, we suggest that implementation-focused research should not only concentrate on technology itself but also integration strategies and policy considerations to optimize the effects of AI-driven AWE tools.

Process-oriented research on feedback engagement

Recent research primarily examined the effectiveness of AWE on language outcomes and there is little knowledge about how learners process automated feedback (Xue, 2024). In order to enrich the related literature, think-aloud protocols, eye-tracking, and analyses of revision behaviors should be carried out to explain how and why certain students respond effectively, whereas others struggle with AI-generated feedback. Knowledge of such reasoning processes including cognitive, affective, and motivational will continue to be critical in the improvement of AWE tools to gain the maximum educational benefit for students at various proficiency levels.

Taken together, these six research directions will ensure AWE advances in tandem with pedagogical best practices while providing transformative, evidence-based writing support for students from elementary through post-secondary education levels, regardless of their linguistic, cultural, or institutional backgrounds.

Practical implications

The present meta-review provides concrete guidance based on empirical evidence from over 300 primary studies for instructors who seek to optimize AWE implementation. Instructors should implement sequential feedback models that deploy AWE tools from surface-level corrections in grammar and mechanics to deeper feedback to build a stronger cohesion and argumentation. In addition, teachers should also provide guidance as well as scaffolding to help students to regulate and effectively use AWE systems. To maximize the integration of the appropriate scaffoldings, educational context should be considered. Based on the results of the moderator analysis, young learners, who showed a modest benefit from the use of AWE systems, require increased teacher mediation of AWE feedback and explicit instruction in interpreting automated suggestions. Furthermore, our meta-review suggested that 3–9-week period is optimal, which provides sufficient time for multiple revision cycles while maintaining student engagement throughout the process.

The findings of this study are also informative for developers. Given the emerging ethical considerations in the current research debates, developers should address several critical aspects for ethical implementation. First, there should be clear mechanisms for transparency describing how algorithms generate feedback. Second, bias mitigation strategies must address, for example, instead of primarily focus on English, AWE systems can incorporate diverse rhetorical patterns and cultural communication styles from multiple linguistic traditions. Third, privacy safeguards require comprehensive policies governing student writing data with explicit guidelines on data retention periods, third-party sharing restrictions, and algorithmic training applications. Lastly, design principles must maintain appropriate human oversight that positions AWE tools as complementary to rather than replacement for human expertise. In conclusion, although AWE systems are designed to assist learners to improve their writing, AWE developers should also create space for students' creativity and respect their privacy.

Limitations

Generalizability of the meta-review may be inhibited to some extent because most of the AWE systems have been trained mainly on English corpora and do not line up with writing patterns found across other linguistic and cultural communities. In addition, perceptions of automated and human feedback, preferences for direct or indirect correction, and roles of teachers and students in feedback processes can shape effectiveness of AWE beyond Western settings. As a result, estimates of effect and guidance for design may carry over only in part to non-English environments.

Conclusion

In conclusion, it appears that AWE deals well with basic grammar, mechanics, and vocabulary problems, but not as well with higher-order skills like argumentation, coherence, and critical thinking. There have also been reports that peers and teachers have a big impact on how well automatic feedback works. Historical analyses of the technology posit that AWE has progressed from rudimentary rule-based tools into sophisticated AI-driven systems. The section of meta-meta-analyses further reveal that moderate-to-somewhat-strong effect sizes can be reached. In the end, upon the completion of the meta-synthesis study, we would like to conclude that AWE alone is infrequently adequate and call for the establishment of blended models that integrate automated feedback, teacher facilitation, peer interaction, and consideration of cultural and instructional context.

Appendix

See Tables 4 and 5

Table 4 The results of PRISMA checklist and fit statistics

Depth of methodological examination: prisma-anchored evidence

Study	No. of Primary Studies	PRISMA Coverage (%)	Com-pleteness (%)	Outfit MNSQ	Key Methodological Strengths	Critical Methodological Gaps
Chen et al. (2025)	25	81.5	77.8	1.399	• Identified as systematic review & meta-analysis • Full search strategy across multiple databases • PRISMA flow with dual screening • Reporting-bias tests • Clear declarations	• No study-level risk-of-bias appraisal • No certainty grading • No protocol/registration • Partial duplicate extraction • Partial per-study reporting
Nunes et al. (2022)	8	80.8	75	1.46	• Dual screening with Rayyan • PRISMA flow diagram • RoB 2 and ROBINS-I with robvis figures • Systematic bias assessment	• No certainty grading • No publication-bias analysis • Partial search reporting • Partial duplicate extraction
Huang et al. (2024)	22	81.5	70.4	1.418	• PRISMA flow diagram • Explicit eligibility & data items • Random-effects synthesis • Begg and Egger tests	• No study-level risk-of-bias appraisal • No certainty grading • No protocol/registration • Generic data-availability statement
Fleckenstein et al. (2023)	20	77.8	68.5	1.474	• PRISMA flow diagram • Clear effect-size handling • Tree-level/random-effects modeling • Egger testing	• No study-level risk-of-bias appraisal • No certainty grading • No protocol/registration • No reporting/estimation independence
Xue (2024)	29	77.8	61.1	1.472	• Defined inclusion/exclusion criteria • REML with Hartung-Knapp adjustments • Heterogeneity & moderator analyses • Begg and Egger tests	• No PRISMA flow for meta subset • No dual screening for meta subset • No study-level risk-of-bias appraisal • No certainty grading • No protocol/registration
Ngo et al. (2022)	44	74.1	57.4	1.554	• Clear inclusion/exclusion criteria • Reported effect-size formulas • Three-level meta-analysis with moderators	• No study-level risk-of-bias appraisal • No certainty grading • No protocol/registration • No publication-bias tests • No PRISMA diagram

Table 4 (continued)

Depth of methodological examination: prisma-anchored evidence

Study	No. of Primary Studies	PRISMA Coverage (%)	Com-pleteness (%)	Outfit MNSQ	Key Methodological Strengths	Critical Methodological Gaps
Shi and Aryadoust (2024)	83	73.1	59.6	1.582	• Systematic review in title • PRISMA flow with counts • Pre-specified variables • Combined quantitative & narrative synthesis	• No study-level risk-of-bias appraisal • No certainty grading • No protocol/registration • No publication-bias tests • Unclear dual screening independence
Karatay and Karatay (2024)	40	81.8	68.2	1.702	• PRISMA flow diagram • Reported inter-coder reliability • Grounded-theory synthesis	• No qualitative risk-of-bias appraisal • No CERQual/certainty assessment • No protocol/registration • Reported search details
Fu et al. (2024)	48	57.7	49.1	1.804	• Clear eligibility & sources • Predefined coding • Broad categorical synthesis • Funding & COI statements	• No PRISMA flow or dual screening • No study-level risk-of-bias appraisal • No systematic bias assessment • No certainty grading • No protocol/registration
Cheung (2015)	4	36.4	31.8	1.89	• Stated rationale & research questions • Basic eligibility & sources	• No PRISMA flow • No search strategy • No data extraction • No study-level risk-of-bias appraisal • No certainty grading • Minimal study characteristics
Zhang (2021)	0	21.1	18.4	1.94	• Stated rationale • Funding statement	• No eligibility/database specification • No search strategy • No data extraction • No study-level risk-of-bias appraisal • No certainty grading • No per-study results

Table 5 Codebook of the study

Code ID	Code Definition	Evidence from Reviews	Analysis Application	Figure/ Output	Methodological Process	Co-occurrence Pattern
TE1	**Grammar-focused feedback dominance** Early AWE concentratec on grammar and mecha with generic comments	• *Cheung (2015) describes AWE feedback as "abstract, vague, unspecific, formulaic, [and] repetitive," contrasts focus on grammar/mechanics with teachers' fuller guidance.* • *Fu et al. (2024) note that AWE feedback is often generic or indirect and may miss discourse-level issues.*	Temporal clustering for Evolution Timeline	Figure 2: First Generation (2000–2010)	Chronological analysis of tool capabilities across time periods	Appeared with time markers in 6/11 reviews
TE2	**LLM transparency & hallucination risks** AI-powered tools introdu concerns about opaque decisions and errors	• *Chen et al. (2025) discuss concerns about fairness, accountability, transparency, and ethics of AI systems used for assessment.* • *Karatay & Karatay (2024) report loss of trust when AWE feedback is inaccurate.*	Recent technology challenges identification	Figure 2: AI Generation (2022–2024) Figure 6: Emerging Gaps	Temporal frequency analysis for emerging themes	Only in 2022+ reviews, co-occurs with ethical codes
EE1	**Stronger gains on surface features** Consistent positive effec on grammar/spelling vs. limited impact on cohere	• *Cheung (2015) highlights improvement on surface-level accuracy while pointing out weaker support for higher-order qualities.* • *Shi & Aryadoust (2024) summarize AWE most effective for correctness-oriented feedback.*	Effect size categorization and limitation analysis	Figure 4: Effect Size Analysis Moderator Discussions	Meta-analysis synthesis of outcome patterns	Consistent across 8/11 reviews, paired with limitation codes
EE2	**Duration sweet spot** Medium interventions (3–9 weeks) show large gains than very short/lor	• *Xue (2024) shows duration as significant moderator; 1–2 month window outperforms.* • *Fleckenstein et al. (2023) report moderation by length.* • *Chen et al. (2025) find largest effects for 1–8 week implementations.*	Moderator analysis synthesis	Figure 5: Duration Effects	Cross-study moderator pattern identification	Found in 4/5 meta-analyses, consistent pattern
IF1	**Teacher-mediated AWE improves outcomes** Better results when teac help interpret automatec suggestions	• *Nunes et al. (2021) argue AWE should complement rather than replace teachers.* • *Shi & Aryadoust (2024) affirm central role of teachers.* • *Chen et al. (2025) recommend teachers guide students in K–12 settings.*	Implementation strategy identification	Figure 3: Pedagogical Integration increase Figure 7: High-priority solutions	Solution-focused thematic analysis	Co-occurs with student difficulty codes in 7/11 reviews
IF2	**Educational level moderates effects** University settings show larger effects than primary/secondary	• *Fleckenstein et al. (2023) show larger gains at tertiary level.* • *Ngo et al. (2022) report stronger effects for undergraduates.* • *Chen et al. (2025) cite tertiary-level advantages.*	Moderator analysis synthesis	Figure 5: Educational Level Effects	Cross-study moderator pattern extraction	Quantitative pattern in 4/5 meta-analyses
SE1	**Difficulty interpreting automated feedback** Students struggle to ma sense of automated suggestions without guic	• *Cheung (2015) notes students' difficulties understanding automated suggestions.* • *Karatay & Karatay (2024) connect low AWE accuracy to reduced user trust.*	Persistent challenge identification	Figure 6: Student Feedback (Gap) Figure 3: Student Perceptions decline	Cross-review frequency analysis of user experience themes	Most frequent gap – appeared in all 6 systematic reviews
SE2	**Mixed affective responses** Motivation can rise while anxiety drops, yet experiences vary	• *Huang et al. (2024) quantify positive effects on motivation and decreased writing anxiety along with performance gains.*	Affective outcome synthesis	Affective outcomes discussion	Effect size extraction from meta-analyses	Consistent in meta-analyses, variable in systematic reviews
RL1	**Limited longitudinal evidence** Few studies track sustai writing development afte AWE use	• *Nunes et al. (2021) call for longitudinal work to examine sustained improvement and transfer.* • *Shi & Aryadoust (2024) echo need for long-term evidence.*	Research gap identification	Figure 6: Long-term Effects (Gap) Figure 7: Y-axis priority	Gap frequency analysis across review conclusions	Mentioned as limitation in 5/6 systematic reviews
RL2	**Cross-linguistic generalizability concerns** Performance across languages/cultural conte uncertain	• *Chen et al. (2025) show 88% of studies target English with only isolated cases in Chinese, Italian, Finnish.* • *Huang et al. (2024) highlight cultural and contextual considerations.*	Equity and transferability gap identification	Figure 6: Diverse Linguistic Contexts Figure 7: Cultural research priority	Limitation and future direction thematic synthesis	Growing concern – 2/6 early reviews vs 4/5 recent reviews

Analytical Process Summary:

1) **Temporal Analysis:** TE codes revealed technology evolution patterns

2) **Frequency Analysis:** SE1 persistence across all systematic reviews

3) **Co-occurrence Analysis:** IF1 + SE1 pairing informed pedagogical solutions

4) **Meta-synthesis:** EE codes aggregated effect size patterns

5) **Gap-Solution Mapping:** RL codes (gaps) × IF/EE codes (solutions) → Priority matrix

Acknowledgements The authors would like to express their utmost gratitude to the esteemed researchers of the primary studies included in this meta-analysis, whose invaluable contributions have greatly enriched this research. Sincere appreciation is also extended to the anonymous reviewers and editors of Educational Technology Research and Development journal for their valuable and constructive comments, which have played a crucial role in enhancing the quality and rigor of this study.

Funding Open access funding provided by National Taiwan University of Science and Technology. This study is supported in part by National Science and Technology Council under contract number NSTC-112–2410-H-011–011 and NSTC 113–2410-H-011–016.

Data availability The datasets analyzed in the current meta-review are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors have no conflict of interest.

Research involving human participants In this meta-review, we gathered data from an array of previously published studies. Before initiating the analysis, we made sure to de-identify all data, upholding the privacy of individuals. Our research approach focused on minimizing potential risks to human participants.

Informed Consent Since our study did not involve direct human participation, the requirement for informed consent was not applicable. We diligently adhered to relevant ethical guidelines and can confirm that all necessary approvals were secured, allowing our study to contribute to the growing body of knowledge.

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Asterisk (*) indicates the included reviews in the present meta-review

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